



Genetic susceptibility to respiratory health effects from outdoor air pollution: a structured narrative review

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Shareable abstract (@ERSpublications)

A synthesis of 48 air-pollution GxE studies found 89 significant SNPs showed GST/TLR variants consistently amplify pollution-related lung function decline and airway inflammation, calling for multi-ethnic studies and multi-omics for precision care. <https://bit.ly/4cBypIY>

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Abstract

Outdoor air pollution exposure is a well-established driver of respiratory disease, while growing evidence highlights an important role for genetic susceptibility. Building on this expanding body of work, our review synthesises findings across studies and provides an integrated overview. Here, we present a structured literature review summarising study characteristics and statistically significant single nucleotide polymorphisms (SNPs) reported in gene–environment interaction studies of air pollution related respiratory outcomes. By consolidating this evidence, the review clarifies current findings and supports more rigorous, evidence-based risk stratification in future observational and intervention studies. We included 48 peer-reviewed studies in humans published between 2014 and 2024 that examined gene–environment interactions involving SNPs, outdoor air pollution exposure and respiratory outcomes. Studies primarily evaluating particulate matter with aerodynamic diameter $<2.5\ \mu\text{m}$ and $<10\ \mu\text{m}$, and nitrogen dioxide included respiratory end-points, such as asthma, COPD and lung-function measures. The majority of analyses used regression-based methods, supplemented by mixed-effects and generalised estimating equation models. Candidate gene studies (particularly in the glutathione S-transferase (GST) family) along with polygenic risk scores and genome-wide interaction approaches identified 89 SNPs across 53 genes. Replicated variants involved pathways in oxidative stress detoxification (*e.g.* rs1695 (*GSTP1*) and rs2266637 (*GSTT1*)) and inflammatory responses (*e.g.* rs1800629 (*TNF*), rs3804099 (*TLR2*) and rs848 (*IL13*)), consistently modifying pollution-related lung function decline and airway inflammation. By consolidating key genes, study designs and analytic methods, this review provides a curated SNP list to inform future genetic risk stratification. We highlight the need for studies in ancestrally diverse populations, controlled human exposure designs, and multi-omics approaches to strengthen mechanistic understanding of gene–environment interactions in respiratory health.

Introduction

Air pollution remains a major contributor to public health outcomes in recent global disease burden studies [1]. Outdoor air pollution exposure is associated with increased disability-adjusted life years, reduced life expectancy, and higher all-cause mortality rates, and is related to a wide range of noncommunicable conditions, including respiratory and cardiovascular diseases [1–3]. Outdoor air pollution commonly contains particulate matter, ozone (O_3), nitrogen oxides (NO_x), carbon monoxide (CO), other gaseous or organic compounds, and heavy metals [4]. Among these, fine particulate matter, which consists of particles that are $<2.5\ \mu\text{m}$ in diameter ($\text{PM}_{2.5}$), is the most widely studied due to its diverse sources and strong links to adverse health effects.

Anthropogenic sources of $\text{PM}_{2.5}$ are primarily traffic-related air pollution (TRAP), industrial emissions, and agriculture [5]. Natural events, such as wildfires and volcanic eruptions, also contribute to global



particulate matter [2]. Wildfire smoke has substantially contributed to recent elevated PM_{2.5} concentrations in most western states in the United States of America [6]. A similar pattern was observed in Canada during the record-breaking wildfire season in 2023, with concentrations reaching a maximum of 251.1 µg·m⁻³, and >60 days with daily mean PM_{2.5} levels exceeding regulatory thresholds [7, 8]. Importantly, the health impacts of such prolonged exposures depend not only on PM_{2.5} concentrations, but also on the chemical composition of wildfire smoke particles (WSP), which varies depending on the burned material, and can therefore exhibit differing levels of toxicity to human epithelial cells [9–12]. As climate change intensifies summer heatwaves and droughts, wildfires are expected to increase in frequency, size and duration, particularly in tropical and subtropical regions [13]. Given that wildfire smoke can travel thousands of kilometres [14], this will inevitably increase human exposures across vast regions, as we have seen over the past few years [15, 16].

Sufficient evidence shows that both short- and long-term exposures to outdoor air pollution are associated with the development and exacerbation of respiratory diseases such as asthma, COPD and bronchitis [5, 17]. However, there is substantial interindividual variation in susceptibility to respiratory disease development and exacerbation following outdoor air pollution exposure [18]. To better understand the heterogeneity, gene–environment interaction studies have been increasingly used to investigate whether specific genetic polymorphisms modify the effects of air pollution on respiratory outcomes [5]. For instance, genes in the pathways of detoxification (*e.g.* glutathione S-transferase (*GSTM1*, *GSTT1*)) and innate inflammatory responses (*e.g.* Toll-like receptors (*TLR2*, *TLR4*)) have been studied in candidate gene studies to explore gene–environment interactions in the context of asthma and other adverse respiratory outcomes [19–21].

Despite a growing number of gene–environment studies in human health, a relatively small number of studies have focused on respiratory function and disease, while the underlying biological mechanisms remain incompletely understood [22, 23]. Meanwhile, rising interest in precision medicine in clinical settings has been further extended to the field of precision environmental health [24, 25]. Clinical translation of gene–environment research is nascent, but advancing. Polygenic risk scores (PRS) have moved closest to clinical implementation, with studies demonstrating their feasibility in routine care settings through standardised assay workflows and integration into electronic health records [26]. HAO *et al.* [26] showed that PRS-guided screening altered clinical decision-making in breast, colorectal and prostate cancer, as well as atrial fibrillation, coronary artery disease and type 2 diabetes. Another example from LEBRETT *et al.* [27] demonstrated similar effects in lung cancer. Both studies provide proof-of-concept that genomic risk stratification can meaningfully inform patient management. While direct clinical application in respiratory disease remains limited, these findings establish a plausible translational pathway and underscore the importance of developing similarly robust gene–environment models in this field. There are more advances and affordable choices in deeper sequencing technologies [28, 29] and artificial intelligence driven analytical and predictive approaches [30, 31] over the past decade. Researchers now have unprecedented tools to perform deep characterisation of genetic and molecular responses in the respiratory system through gene–environment interaction studies.

However, the most recent comprehensive synthesis of gene–environment interactions related to outdoor air pollution and respiratory health was published in 2014 [5]. To address this gap, we conducted this structured narrative review of peer-reviewed human gene–environment interaction studies published between 2014 and 2024.

This review summarises current knowledge on how genetic variations modify respiratory outcomes in the context of ambient air pollution exposure, compares study designs and statistical methods used across studies, and identifies single-nucleotide polymorphisms (SNPs) with strong associations in gene–environment interactions. Integrating these findings provides an up-to-date overview of genetic susceptibility to air pollution-related respiratory effects, offering a valuable foundation for future gene–environment studies, multi-omics investigation, and prevention-oriented precision health research [32].

Methods

Inclusion criteria

We conducted a structured narrative literature review search, guided by Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines [33], to enhance reproducibility. Three researchers evaluated peer-reviewed human studies published in English from January 2014 to December 2024, that investigated gene–environment interactions involving genetic polymorphisms, outdoor air pollution, and respiratory diseases or lung function impairments. Records were managed in Covidence. Y. Meng and C. Chen performed the title/abstract screening independently; Y. Meng and J.S. Lim conducted the

full-text screening. Disagreements were resolved by discussion and consensus. This review builds upon a previous synthesis by VAWDA *et al.* [5] in 2014, and focuses on the publications thereafter. Eligible studies were required to examine how genetic susceptibility modifies the effects of outdoor air pollution on the risk of respiratory diseases or lung function impairment, among all age groups. Outdoor air pollution encompassed all exposures that were in an outdoor setting, including particulate matter, gaseous pollutants, or organic compounds, and heavy metals. Genetic susceptibility measurements included SNPs or other genetic variants, such as insertions/deletions, haplotypes and copy number variants. Health outcomes included respiratory diseases, mainly in the airways (asthma, COPD, cough, chronic rhinosinusitis and bronchitis), and measurements of lung function and airway inflammation (forced expiratory volume in 1 s (FEV_1), forced vital capacity (FVC), fractional exhaled nitric oxide (F_{ENO}) and inflammatory biomarkers). All relevant observational or interventional study designs were included, such as cohort studies, case-control studies, controlled human exposure trials, and genome-wide interaction studies (GWIS). This review focused on gene-environment interactions in observational or controlled human exposure studies and aimed to extract SNPs that showed significant associations in these studies. Thus, we excluded Mendelian randomisation, *in vitro*/cell-based, animal, non-English, review articles, articles without available full text, and studies of non-outdoor exposures (*e.g.* indoor cooking, tobacco smoke) or end-points other than the aforementioned respiratory diseases. Respiratory infections were excluded. Chronic or subchronic respiratory conditions may be influenced by cumulative environmental exposures through relatively stable molecular mechanisms. In contrast, respiratory infections are typically acute, pathogen-driven conditions shaped by dynamic host immune responses and other time-varying factors, which introduce heterogeneity unrelated to air pollution exposure [34]. The screening process is summarised in figure 1.

Given the heterogeneity of study designs, exposure metrics, and health outcome measures, we performed a thematic narrative synthesis instead of quantitative pooling. This effort should help motivate and support a meta-review in the future, with careful selection and categorisation of publications, when more relevant studies with comparable designs become available.

Search strategy

We performed a structured literature search across four major databases: MEDLINE, Embase, Cochrane Library, and Web of Science. The search strategy was structured around four conceptual domains: genetic susceptibility, air pollution exposure, respiratory health outcomes, and study type. We used a combination of controlled vocabulary and free-text terms to maximise sensitivity. The detailed search methods can be found in supplementary table S5.

Search criteria:

- 1) Gene or synonyms: genetic susceptibility, gene environment, genetics, allele, single-nucleotide polymorphism, polymorphism, variants.
- 2) Exposure or synonyms: exposure, outdoor air pollution, particulate matter, PM, ozone, NO_x , sulfur dioxide (SO_2), diesel, wildfire smoke, wildfire, wood smoke.
- 3) Health outcome or synonyms: airway diseases, airway inflammation, oxidative, asthma, chronic obstructive pulmonary disease, COPD, respiratory disease, respiratory health, lung function.
- 4) Study type: observational or interventional study designs that analyse the association between genetic predisposition, outdoor air pollution, and airway diseases in humans. We included cohort studies, case-control studies, crossover controlled human exposure studies, and GWIS. In this review, we excluded Mendelian randomisation, *in vitro* studies, and animal studies.

Results

48 studies were included in this review, each examining gene-environment interactions involving genetic polymorphisms, outdoor air pollution, and respiratory outcomes (figure 1). The number of studies on this topic has increased substantially since 2014, compared with only eight studies identified in the previous review published in 2014 [5]. Full details of the included publications are provided in supplementary table S1. Results are presented in a thematic narrative synthesis structure to summarise and analyse characteristics of the included studies, highlighting the gene-environment interaction findings. The statistically significant SNPs identified are detailed supplementary table S2.

Study characteristics and population distribution

The included studies spanned diverse geographic regions (Europe, North America, East Asia, Africa and Australia) and included participants from various age groups and ethnic backgrounds. Despite this diversity, there was disproportionate representation of regions, with 36 out of 48 included studies focusing on populations of European ancestry. There were 10 studies in East Asia, and two from Africa. The limited

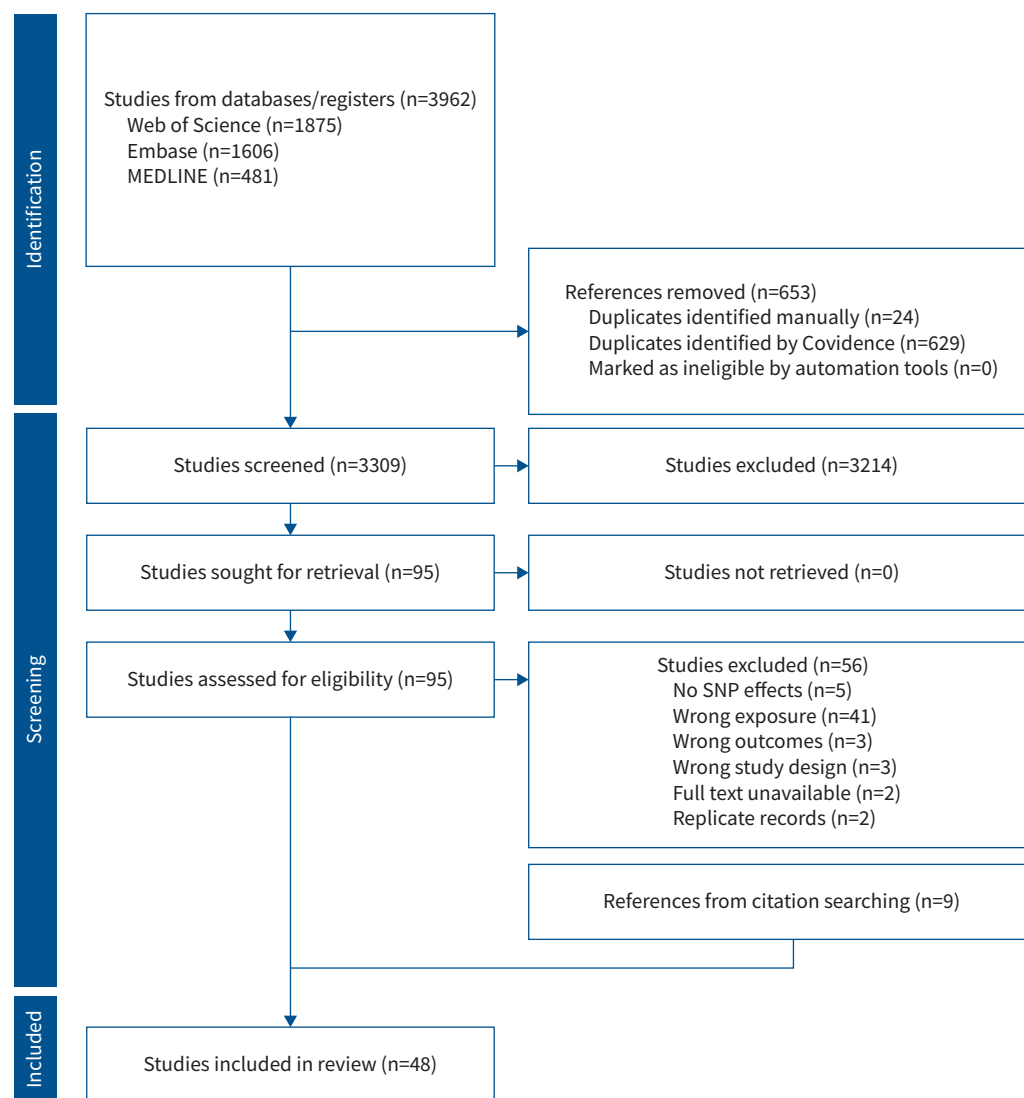


FIGURE 1 Process from study identification to inclusion.

number of studies from East Asia and Africa can be attributed to the fact that most cohorts were recruited in North America, Australia or Europe. There were a few cohorts in Korea [19, 35] and in Taiwan [36]. Three studies [20, 37, 38] combined multicontinental cohorts.

The varying frequency of genetic variants in diverse ethnic groups could affect susceptibility to certain diseases, including asthma [38], which was also found in the included studies. One GWIS conducted a meta-analysis of genetic susceptibility to asthma by combining multi-ancestry cohorts [39]. There were 49 out of 179 loci that did not reach genome-wide significance in a European-only meta-analysis, and 13 novel loci that had an index variant higher in frequency in the non-European ancestry group, compared with the European ancestry group [39]. Another study compared the differences in genetic susceptibility between German and Japanese elderly women. Although no significant gene–environment effects were reported on asthma outcomes, the authors observed a protective trend in German women, but an opposite adverse effect in Japanese women. These inconsistent findings highlighted that more attention to ethnicity-specific genetic susceptibility may be needed [26].

Figure 2 summarises the temporal, regional, and demographic distributions of the included studies, and the study designs. The included studies differed in exposure measurements, participant numbers, follow-up methods, and encompassed prospective and retrospective cohort studies, case–control designs, cross-sectional studies, panel studies, crossover controlled human exposure experiments, and GWIS.

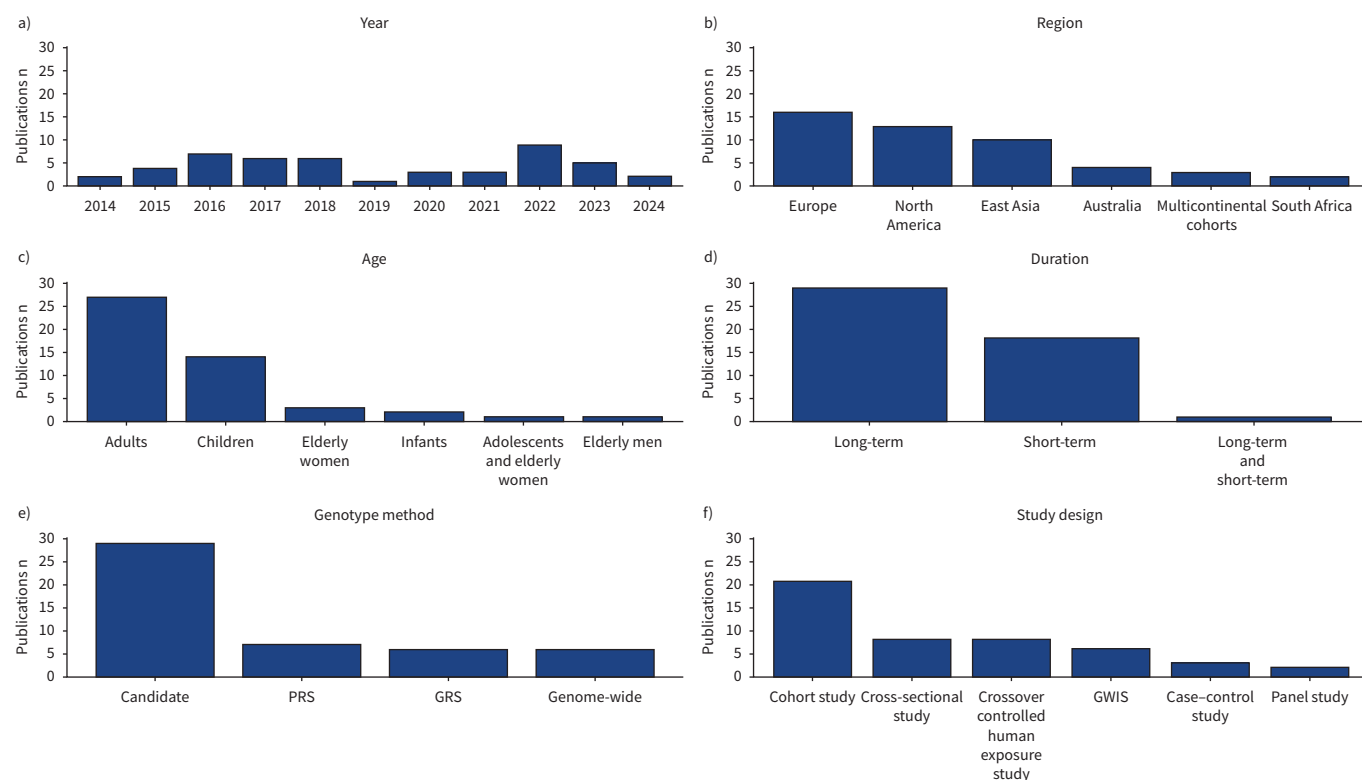


FIGURE 2 Publications per year, region, age, duration, genotype method, and study design. For clarity, we present results separately for adolescents, general adult populations, and elderly women or men. Although older populations are a subgroup of adults, we highlight them due to their distinct risk profile as reported in prior studies. PRS: polygenic risk score; GRS: genetic risk score; GWIS: genome-wide interaction study.

Exposure type, health end-points and measurements

The included studies assessed a wide range of outdoor air pollutants, with particulate matter and NO_2 being most frequently investigated. Particulate matter was represented in a variety of forms: $\text{PM}_{2.5}$, PM_{10} , diesel exhaust particles (DEP), and woodsmoke particles. Gaseous pollutants included NO_2 , NO, O_3 , CO and SO_2 . Several studies focused on polycyclic aromatic hydrocarbons, such as benzo[a]pyrene.

Studies with smaller sample sizes primarily assessed short-term exposures, particularly through crossover controlled human exposures [40–47] and panel designs [48, 49]. In controlled exposure studies, each participant was exposed to a standard concentration of DEP or WSP in a controlled environment. DEP exposure levels in the experimental booth were standardised to $\sim 300 \mu\text{g}\cdot\text{m}^{-3}$ of $\text{PM}_{2.5}$, along with CO, NO and NO_2 [45], while $>190 \mu\text{g}\cdot\text{m}^{-3}$ of $\text{PM}_{2.5}$ was used for the WSP exposure studies [47]. One panel study and one case-control study used data from real-time air monitoring stations or devices [48, 50]. Another panel study used portable air samplers [51] to assess individual-level exposures.

For long-term exposures, 33 out of 38 cohort studies, cross-sectional studies, and GWIS used ArcGIS tools (land-use regression models or the inverse distance weighting method [19, 35, 36, 38, 50, 52–54]) to interpolate the level of pollutants from fixed-site monitoring data (e.g. the United States Environmental Protection Agency database [55]). Three studies used proximity to roads and farms to estimate real-world exposure to TRAP [56, 57] or microbes [58]. These studies typically had larger populations and longer follow-up durations.

Multiple health end-points were considered to represent respiratory diseases, including disease risk, lung function impairments, and inflammation, as well as lung function-related biomarkers. Disease risk was typically assessed using measures such as odds ratio, relative risk, prevalence, or hazard ratio, comparing exposed and nonexposed groups for the relevant pollutants [27, 43, 44]. Lung function impairment was commonly evaluated through spirometry (FEV_1 , FVC) and F_{ENO} testing. Spirometry is widely used in diagnosing respiratory diseases such as asthma and COPD [59, 60]. F_{ENO} , spirometry, and other noninvasive measurements can be repeated at different time points, as is normal in crossover studies [41, 47].

In addition to noninvasive assessments, primary respiratory epithelial samples can be obtained through methods such as bronchoscopy or induced sputum collection. These procedures yield samples such as bronchial brushings, bronchial wash, bronchoalveolar lavage, and sputum, which can be used to extract DNA/RNA, assess cell differentiation, profile cytokines, and measure sputum neutrophil percentages [36, 38]. Researchers either chose a single or combined method of noninvasive and invasive collection to obtain samples and health end-points based on their hypothesis and research plans.

Counts of publications by environmental exposure *versus* health outcomes are shown in figure 3a. Studies with larger populations often had more than one exposure of interest measured and more than one health end-point. The most studied exposure *versus* health end-point was lung function and/or airway inflammation *versus* particulate matter (39.6% of included publications), followed by NO₂ (27.1% of included publications). Among these, three studies assessed airway inflammation exclusively using *F*_{ENO} as the clinical end-point [49, 61, 62].

Study design and statistical methods

Six major types of study design were represented in the included publications: cohort, case-control, cross-sectional, panel, crossover controlled human exposure, and GWIS. Two studies were originally classified as genome-wide association studies (GWAS), but as they performed gene-environment interaction analyses, they were reclassified as GWIS [55, 63]. Supplementary table S3 summarises the key features, strengths and limitations of the study designs employed in the included studies.

Regression-based models were the most frequently applied statistical methods in gene-environment analyses, with linear and logistic regression models being the predominant choices. Mixed-effects models (including linear and generalised additive mixed models) were primarily applied in studies that had repeated within-participant measurements of exposures at different time points. Generalised estimating equation analysis was performed to assess repeated measurements, either independently or alongside regression-based models, in several studies [43, 64–67]. Two studies applied a nonlinear dose-response [52] method or Bayesian models [35] to analyse gene-environment interactions. Supplementary table S4

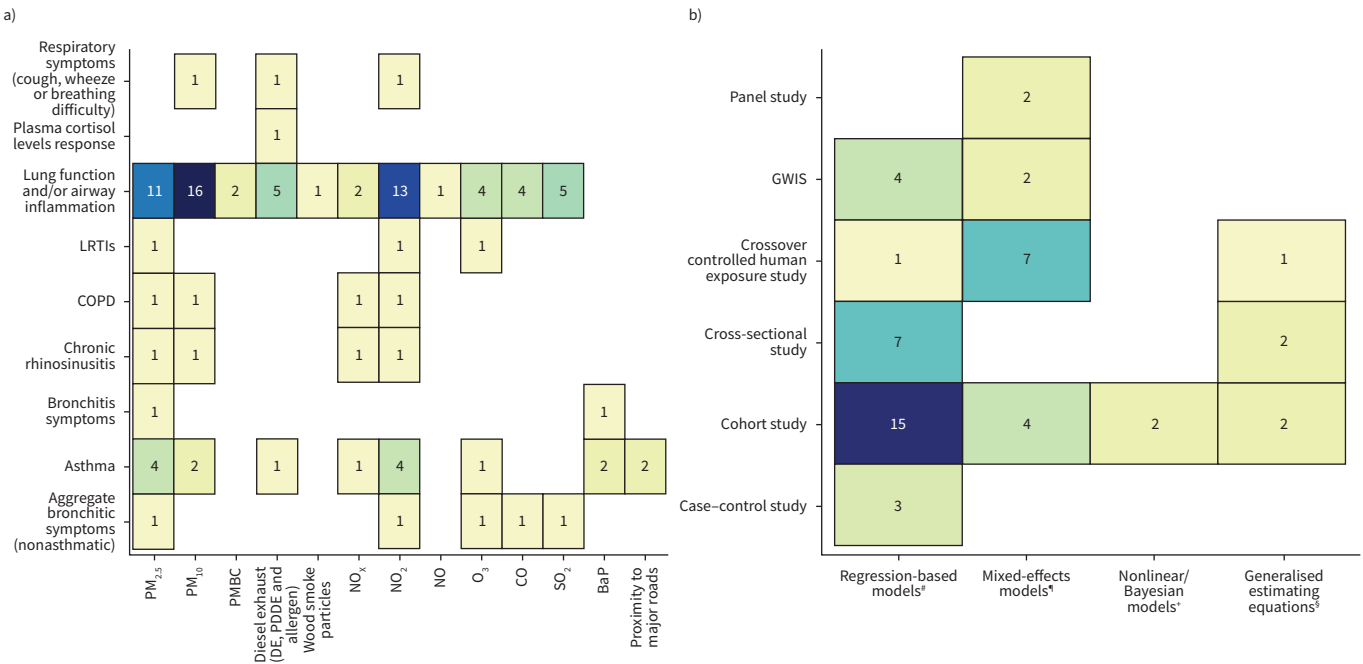


FIGURE 3 Publication numbers by a) exposure type and health outcomes, and b) study design and statistical methods used. LRTI: lower respiratory tract infection; PM_{2.5/10}: particulate matter with aerodynamic diameter ≤2.5 μm or ≤10 μm; PMBC: particulate matter-bound black carbon; DE: diesel exhaust; PDDE: particle-depleted diesel exhaust; NO_x: nitrogen oxides; NO₂: nitrogen dioxide; NO: nitric oxide; O₃: ozone; CO: carbon monoxide; SO₂: sulfur dioxide; BaP: benzo[a]pyrene; GWIS: genome-wide interaction study. ^a: logistic, linear, general linear, Poisson, Cox-proportional hazard, quantile, or least absolute shrinkage and selection operator regression models; ^a: linear mixed effects, generalised additive mixed models, and generalised mixed-effects models; ^a: Bayesian distributed lag interaction models, nonlinear dose-response analysis; ^a: generalised estimating equations were normally combined with regression models to analyse the effects of different time points.

summarises the commonly used statistical methods in the studies included in this review, while figure 3b summarises publication numbers by study design and the statistical methods used.

Gene–environment interaction findings

40 out of the 48 included studies reported at least one statistically significant gene–environment interaction, while the remaining eight studies reported null [38, 50, 51, 54, 63, 68, 69] or suggestive associations [65]. Across all included studies, there were 89 SNPs across 53 unique genes that reached statistical significance (excluding the two null polymorphisms of *GSMT1* and *GSTT1*, and SNPs in the PRS models). Details of these 89 SNPs can be found in supplementary table S2. Most of these significant genes fell into key biological pathways, including immune and inflammatory responses (e.g. *TLRs*, *TNF*, *TOLLIP*), oxidative stress or detoxification (e.g. *GST* family, *NFE2L2*, *SOD2*), signalling and sensing (e.g. *TRPV1*, *TRPA1*), and epithelial regulation (e.g. *CFTR*, *SFTPA*, *MUC5B*). Genes were annotated and grouped by function using the Single Nucleotide Polymorphism Database [70], and visualised in figure 4 using Cytoscape [71].

The most frequently studied candidate genes were in the *GST* family. The null genotypes of *GSTM1* and *GSTT1* were associated with an increased risk of respiratory disease symptoms [19, 57, 72] or decreased lung function [45–49, 61, 64, 73, 74], following exposure to outdoor air pollution, including particulate matter, DEP, WSP, NO₂, O₃, SO₂, CO or proximity to major roads. The same was found for the SNPs in *GSTs*, such as rs762803, rs8191438, rs1695 and rs1138272 in *GSTP1*, and rs2266637 in *GSTT1*. Another key group of candidate genes was involved in immune and inflammatory response pathways, such as rs1800629 in *TNF*, rs2569190 in *CD14*, rs3804099 in *TLR2*, and rs848 in *IL13*, etc. This result also confirmed the synthesis conducted by VAWDA *et al.* [5], that variants in inflammatory and immune responses are modifying the interaction between respiratory diseases and air pollution.

In addition to the candidate genes, genetic risk scores (GRS) or PRS were often used to quantify potential susceptibility. GRS were derived from manually selected SNPs based on the literature or

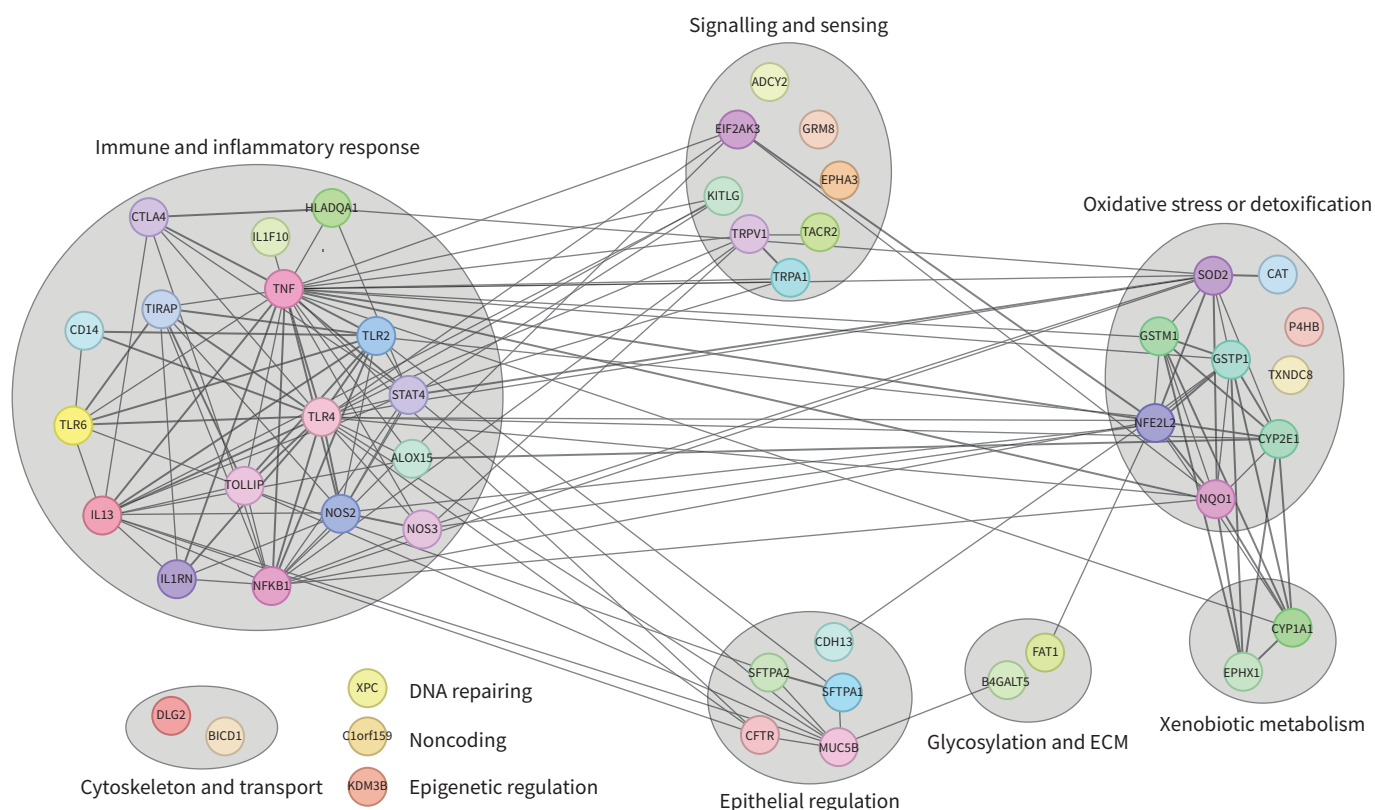


FIGURE 4 Network of genes exhibiting significant gene–environment interactions in the included studies, clustered by key cellular processes. Note: 89 single nucleotide polymorphisms in 53 unique genes reached a level of significance and involved major pathways of inflammation, oxidative stress, signalling, sensing or epithelial regulation. Genes were annotated and grouped using the information in the Single Nucleotide Polymorphism Database, and the figure was created in Cytoscape. We were unable to plot five genes in noncoding areas. ECM: extracellular matrix.

specific pathways. Overall, studies using GRS, particularly those based on selected SNPs in inflammatory response pathways, consistently found that higher scores were associated with greater cough counts, heightened inflammatory responses, increased risk of respiratory disease or symptoms, and declines in lung function following exposure to diesel exhaust particles or higher concentrations of air pollution [36, 40, 42, 43, 56, 58, 75–78]. PRS were constructed using weighted or unweighted sums of risk alleles [38, 55] for SNPs that were identified in GWAS [38, 52, 53, 62, 68, 79]. Four studies found associations in the gene–environment interactions among certain subgroups, and had features of dose-dependent and additive effects [52, 53, 62, 79]. The remaining two studies did not report a significant association in the main PRS model [38, 68].

GWIS analysis identified novel loci associated with lung function decline and found significant associations in the gene–environment interaction analyses, including with rs4143882, rs6886921, and rs727432 in *ADCY2* (with NO₂), rs17284098 and rs2325934 in *CDH13* (with PM₁₀), rs12312730 in *BICD1* (with PM₁₀), and rs58475486 in *KITLG* (with SO₂) [37, 55, 80, 81]. Among all these effects, *KITLG* appeared to be specific to African American children, while *EPHA3* and *TXNDC8* were specific to populations of European ancestry, underscoring the importance of accounting for genetic ancestry in gene–environment studies [73, 82].

Discussion

Comparing with the review article in 2014 [5], this narrative review demonstrates the rapid growth of gene–environment studies that examine interactions between air pollution and respiratory health. We also observed methodological advances, with ~20% of the included studies applying more rigorous analytical frameworks, including at least one of: Hardy–Weinberg equilibrium test, linkage disequilibrium, or expression quantitative trait locus analysis [55, 56, 78, 82]. These approaches are a positive step, along with advances in sequencing technologies, strengthened robustness and standardisation of methodology, association reporting and improvements in biological interpretation.

This narrative review included 48 publications discussing gene–environment interactions and how genetic variants related to health outcomes across diverse age groups and multiple continents. Outdoor air pollution exposures during prenatal development, childhood and adulthood were consistently associated with elevated risk of asthma, COPD and chronic rhinosinusitis, increasing airway inflammation, and reduced lung function independently or dependently of genetic susceptibility [35]. This association reflects multiple pathways, including pollutant-induced oxidative stress (*e.g.* GST family), inflammatory responses (*e.g.* TLRs), and epithelial barrier dysregulation (*e.g.* *SFTPA*).

The prenatal and postnatal periods are critical windows for airway development, during which exposure to air pollution can lead to adverse respiratory effects and long-term morbidity that may persist into adulthood, with genetic predisposition potentially modifying these associations [57, 77]. Long-term outdoor air pollution exposures were found to be linked with chronic airway remodelling, increased pro-inflammatory cytokine release, and progressive declines in lung function [36, 52, 57]. In contrast, short-term exposures, examined through crossover controlled human exposure and panel studies, provided mechanistic evidence of acute inflammatory and airway responses [41, 44, 45, 48, 49].

Two panel studies illustrated the variability of particulate matter exposure and lung function. One study found that *GSTM1* null carriers exhibited elevated inflammatory cytokine levels and reduced lung function, with positive associations observed for short-term exposure to PM_{2.5} or PM₁₀ in healthy undergraduates, but inverse relationships with gaseous pollutants [48]. In contrast, the other study reported a negative association between lung function and increased PM_{2.5} levels [49]. These inconsistencies underscore the importance of assessing dose–response relationships, particle deposition mechanisms, and genetic susceptibility to the impacts of short-term exposure on the airways [44, 48].

Notably, two studies reported no significant associations for either gene or exposure alone but identified significant effects in gene–environment interaction models. One study found five SNPs in the *MUC5B* and *TOLLIP* region that had a significant interaction with NO₂ exposure, which increases respiratory symptoms in the first year of life [77]. The other observed a strong association of lower risk of early-onset asthma with childhood farm exposure for participants carrying the T allele in *TLR6*-rs1039559 and the C allele in *TLR6*-rs5743810 [58]. This highlights the critical value of including genetic susceptibility in exposure–outcome analyses to explain interindividual variability in respiratory disease risk.

Moreover, despite the increasing frequency and duration of wildfires and existing evidence of adverse respiratory system effects due to inhalation of WSP, only one out of the 48 included studies analysed how

GSTM1 modified the interaction of WSP with lung function decline [47]. Further research is needed to address this gap, explore the heterogeneity of WSP composition, and inform effective policy and prevention strategies.

Gene–environment interaction

GST family members were the most frequently studied candidate genes and were consistently associated with increased risk of asthma and lower respiratory tract infections, heightened inflammation, and lung function decline. *GSTM1* and *GSTT1* null genotypes showed particularly strong associations with adverse respiratory outcomes in response to particulate matter, DEP, WSP and other gaseous pollutants, compared to wild-type carriers [46, 48]. One study also reported a higher prevalence of these null genotypes in the GST family in individuals with COPD, compared to the control group [83]. Sufficient evidence highlights the importance of polymorphisms in GST family genes in respiratory diseases corresponding to air pollution. These findings support existing evidence that the GST family plays a key role in detoxification in response to air pollutants [19, 45, 47].

Beyond oxidative stress pathways, several studies that used GRS based on selected SNPs focusing on immune and inflammatory responses, found that higher GRS levels were associated with increased risk of respiratory disease or symptoms, inflammation, and lung function decline in response to outdoor air pollution [36, 40, 56, 58, 77, 78].

PRS and GWIS offered larger-scale analyses, but yielded a slightly higher proportion of null findings compared with the candidate genes method. The PRS and GWIS models were either derived from GWAS studies or conducted through a genome-wide analysis. In general, additive effects of higher PRS, air pollution, and unfavourable lifestyles were associated with a higher risk of chronic rhinosinusitis or larger lung function declines. Two out of seven PRS studies did not identify a significant gene–environment interaction [38, 68]. Five out of six GWIS studies identified at least one locus and found a significant association in gene–environment analysis [37, 55, 80–82].

In contrast to the adverse effects, some studies reported protective associations of several genes in the gene–environment interactions. One study found that carrying certain SNP alleles of *TLR6* could lower the risk of early-onset asthma in combination with childhood farm exposure [58]. Another GWIS study showed the protective effects of rs58475486 in the *KITLG* gene on lung function in African American children interacting with the past year of SO₂ exposures. There were some inconsistencies among the results, and the heterogeneity of race in the gene–environment interaction analyses highlights the need to involve multiple ethnicities in research.

The mixed findings in this review might prompt a methodological discussion. Candidate gene approaches grounded on biological evidence [84] offer greater statistical power [85] and are mechanistically interpretable, but need to cover significant susceptibility genes or may lead to false positives in small samples, leading to confirmation bias [86]. Conversely, GWIS enables unbiased discovery of novel loci, weak genetic effects, and functional mutations without *a priori* evidence, but multiple comparisons adjustment substantially decreases statistical power [85, 86]. Integrating known disease-associated variants with GWIS-identified loci might be a promising strategy for understanding the role of genetic susceptibility in environment-exposure interactions. Beyond the discussion on methods for evaluating genetic susceptibility, multi-omics analysis might explain the heterogeneity of these findings, by integrating genomic, transcriptomic, and epigenomic data to reveal the underlying mechanisms through which the respiratory system responds to air pollution exposures at the molecular and cellular level [87].

This review has several limitations. We restricted our search to four databases and English-only publications, which may have introduced language bias or excluded relevant studies not indexed in these databases. There is also a strong possibility of underreporting of null or negative results, which contributes to publication bias. This is reflected by the fact that 40 out of the 48 included studies reported at least one statistically significant gene–environment interaction. Another limitation is the substantial variability across gene–environment interaction studies in study design, exposure measurements, outcome definitions, and statistical methods, which restricts the feasibility of conducting a meta-analysis or quantitative synthesis. Subgroup meta-analyses may nevertheless be feasible. Finally, as we only included articles in English, which might have introduced a selection bias, such that this review might underrepresent populations with non-European ancestry. Given the genetic diversity and variability in susceptibility across ethnic groups, future studies need to include more diverse populations to discover population-specific loci for certain health effects.

Conclusion

This structured narrative review demonstrates, from studies published between 2014 and 2024, that genetic variants interact with outdoor air pollution to influence the risk of respiratory diseases, elevate inflammation, and negatively impact lung function. This review summarises study design, analytical methods, exposure, and health end-point measures, highlighting both consistent and inconsistent gene–environment interaction findings, and synthesises a list of SNP shown to significantly modify risk. This review builds the evidence base for broad application in future gene–environment studies, multi-omics investigation, and incorporating environmental exposures within the spectrum of precision health applications. gene–environment interactions are modified through major pathways such as the GST family gene variants for detoxification, and *TLR* pathway polymorphisms for inflammatory responses. Among all genes, GST family members remain the key candidates in this domain, as confirmed by various studies, exposures, and airway outcomes. More recent PRS and GWIS studies have identified novel loci showing significant associations between air pollution exposures and airway health traits. 89 statistically significant SNPs across 53 genes were extracted in gene–environment interactions, providing a valuable foundation for future gene–environment analyses. Moving forward, studies should prioritise increasing genetic ancestral diversity, investigate additional mechanisms, and integrate multi-omics approaches to advance understanding of gene–environment mechanisms in a wider range of respiratory outcomes.

Points for clinical practice and questions for future research

- This review provides an updated synthesis of genetic susceptibility to ambient air pollution related respiratory effects, helping clinicians and public health practitioners recognise that pollution impacts are heterogeneous across individuals.
- The evidence supports prevention-oriented, precision health approaches that incorporate environmental exposure history with genetic susceptibility signals to guide risk communication, monitoring, and exposure-reduction strategies in patients with respiratory vulnerability.
- Future studies should increase genetic ancestral diversity, test additional biological mechanisms, and integrate multi-omics approaches to clarify gene–environment pathways across a broader range of respiratory outcomes.

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